```
import cv2
from google.colab.patches import cv2_imshow
import numpy as np
import os
```

```
image_files = [
    "/content/smoothening data.png",
    "/content/smoothening data2.png",
    "/content/smoothening data3.png",
    "/content/smoothening data4.png",
    "/content/smoothening data5.png",
]
```

```
output_dir = "/content/filtered_images"
os.makedirs(output_dir, exist_ok=True)

filtered_images = []
```

```
for idx, file in enumerate(image_files, start=1):
    img = cv2.imread(file)
    if img is None:
        print(f"⚠️ Could not read {file}")
        continue

    filtered_img = cv2.GaussianBlur(img, (7, 7), 0)
    filtered_images.append(filtered_img)
```

```
    save_path = os.path.join(output_dir, f"filtered_image_{idx}.png")
    cv2.imwrite(save_path, filtered_img)
    print(f"Saved: {save_path}")
```

```
Saved: /content/filtered_images/filtered_image_5.png
```

```
heights = [img.shape[0] for img in filtered_images]
widths = [img.shape[1] for img in filtered_images]

min_height = min(heights)
min_width = min(widths)


resized_images = [cv2.resize(img, (min_width, int(img.shape[0] * min_width / img.shape[1])))
                  for img in filtered_images]


grid = np.vstack(resized_images)
cv2_imshow(grid)
```